

TECHNICAL DESIGN DOCUMENT

Voice Depression Detector

AI-Powered Acoustic Analysis Demo Application

Real-time Audio Capture · WebRTC Noise Cancellation · Acoustic Feature Analysis

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1.0

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Demo / POC

Classification
Technical

DISCLAIMER: This application is designed for research, education, and demonstration purposes only. It is NOT a clinical diagnostic tool and must not be used as a substitute for professional mental health assessment.

1. Executive Summary

The Voice Depression Detector is a browser-based proof-of-concept (POC) application that captures a user's voice in real time, applies WebRTC-based noise cancellation, extracts acoustic features from the audio signal, and uses a heuristic AI scoring model to indicate whether the voice pattern shows characteristics associated with depressive speech.

The application is delivered as a single self-contained HTML file — requiring no installation, no server, no internet connection after download, and no external dependencies. It runs on any modern web browser on PC (Windows, macOS, Linux) and mobile (Android iOS).

Key Objective

Demonstrate the feasibility of voice-based mental health screening using freely available browser technologies — Web Audio API, WebRTC, and Canvas — with an AI scoring layer, in a zero-installation, cross-platform format suitable for live demonstration.

1.1 Key Capabilities

- Real-time voice capture using the browser's native MediaDevices API
- Automatic noise suppression, echo cancellation, and gain control via WebRTC
- Live animated waveform visualisation rendered on HTML5 Canvas
- Acoustic feature extraction: RMS loudness (dB), fundamental pitch frequency (Hz), speech rate proxy (wpm)
- Heuristic depression likelihood scoring based on low pitch, low loudness, and monotone variability
- Visual result display: depression likelihood bar, positive affect score, feature breakdown, and verdict
- Demo simulation mode requiring no microphone permission — suitable for showcasing on projectors
- Fully responsive design — identical experience on laptop screen and smartphone

2. Application Design

2.1 Design Philosophy

The application was designed around three principles:

- Zero friction deployment — a single .html file that opens by double-click. No npm install, no Python environment, no server setup. The entire application — markup, styles, logic, and assets — is embedded in one file under 15 KB.

1. Cross-platform compatibility — runs identically on Chrome, Edge, Firefox, and Safari across Windows, macOS, Linux, Android, and iOS. All APIs used are W3C standards available in every modern browser.
2. Demo-first UX — the demo simulation mode allows the application to be showcased without a microphone, on a projector, or in environments where mic permissions are restricted. The real-mode and demo-mode produce identical UI flows.

2.2 Architecture Overview

The application is a single-page client-side application with no backend. All processing occurs in the browser using native APIs. The architecture has four layers:

Layer	Technology	Responsibility
Audio Capture	MediaDevices API / getUserMedia()	Access microphone stream with noise processing constraints
Signal Processing	Web Audio API / AudioContext	FFT analysis, RMS computation, waveform rendering
Noise Cancellation	WebRTC built-in constraints	noiseSuppression, echoCancellation, autoGainControl
AI Scoring	JavaScript heuristic model	Acoustic feature weighting → depression likelihood score
Visualisation	HTML5 Canvas + CSS animations	Live waveform, noise bar, metric cards, result display

2.3 User Interface Design

The UI follows a dark-themed, data-dense design language appropriate for a technical demo. Key design decisions:

- Dark background (#0a0e1a) with a subtle green grid pattern conveys a real-time monitoring aesthetic
- Animated ring button provides clear recording state feedback (idle / recording / analysing) without text-heavy instructions
- Live metric cards (duration, dB, Hz, wpm) update in real time during recording to demonstrate active signal processing
- Result card uses colour-coded confidence bars — red/amber for high depression indicators, green for positive affect — for instant visual interpretation
- Disclaimer and tags are permanently visible to ensure no user confuses the demo output with a clinical assessment

3. Development Details

3.1 Programming Languages

Language	Role	Volume
HTML5	Document structure, semantic markup, canvas element	~60 lines
CSS3	Visual styling, animations, responsive layout, dark theme	~280 lines
JavaScript (ES6+)	All application logic — audio capture, signal processing, AI scoring, UI state management	~240 lines

No frameworks, no build tools, no package managers. Pure vanilla HTML/CSS/JS. The only external resource is the Google Fonts import for DM Sans and DM Mono typefaces — the application works offline if this is removed and system fonts are substituted.

3.2 Browser APIs Used

API	Purpose	W3C Standard
MediaDevices.getUserMedia()	Request microphone access with audio processing constraints	Media Capture and Streams
AudioContext / Web Audio API	Create audio processing graph, connect source to analyser node	Web Audio API
AnalyserNode	FFT (Fast Fourier Transform) analysis of audio signal in frequency domain	Web Audio API
Uint8Array / getByteTimeDomainData()	Read time-domain waveform data for waveform rendering	Web Audio API
getByteFrequencyData()	Read frequency-domain data for pitch detection	Web Audio API
HTMLCanvasElement / 2D Context	Real-time waveform rendering at ~60fps via requestAnimationFrame	HTML Living Standard
requestAnimationFrame()	Smooth waveform animation loop synchronised to display refresh rate	HTML Living Standard
devicePixelRatio	High-DPI canvas scaling for sharp rendering on Retina / HiDPI displays	CSSOM View Module

3.3 Noise Cancellation Implementation

Noise cancellation is implemented entirely through WebRTC browser constraints — no external audio processing library is required. When requesting microphone access, the application specifies:

```
noiseSuppression: true // Removes background hiss, fan noise, ambient sound
echoCancellation: true // Cancels microphone feedback and room echo
autoGainControl: true // Normalises input volume for consistent signal level
sampleRate: 44100 // Professional audio quality sample rate
```

These constraints invoke the browser's built-in DSP (Digital Signal Processing) pipeline, which on most platforms routes through the OS audio stack (Windows WASAPI, macOS CoreAudio, or platform WebRTC). The noise cancellation progress bar in the UI is an animated visual indicator — actual suppression occurs continuously at the hardware/OS level from the moment recording starts.

3.4 Acoustic Feature Extraction

3.4.1 RMS Loudness (dB)

Root Mean Square amplitude is computed from the time-domain waveform buffer on each animation frame. The formula converts the 8-bit unsigned PCM values (0–255, centred at 128) to a normalised -1 to $+1$ scale, computes the mean square, takes the square root, then converts to decibels:

```
rms = sqrt( mean( ((sample - 128) / 128)^2 ) )
dB = 20 * log10(rms + 1e-6)
```

The $1e-6$ epsilon prevents $\log(0)$ for silence. Typical speech sits between -20 dB and -8 dB. Values consistently below -25 dB indicate quiet or flat speech, which is a known acoustic correlate of depression.

3.4.2 Fundamental Pitch Frequency (Hz)

Pitch is estimated by identifying the dominant frequency bin in the FFT frequency-domain data, focusing on bins 1–300 (roughly 20 Hz to 6 kHz at 44.1 kHz sample rate). The bin index is converted to Hz using:

```
Hz = binIndex * (sampleRate / fftSize)
```

With `fftSize = 2048` and `sampleRate = 44100`, each bin represents approximately 21.5 Hz. Normal conversational speech sits between 85 Hz (deep male voice) and 300 Hz (high female voice). Consistently low pitch below 150 Hz and reduced pitch variability (monotone) are documented biomarkers of depressive speech in acoustic research.

3.4.3 Speech Rate Proxy

A proxy speech rate in words-per-minute is derived from the RMS amplitude using a linear scaling heuristic. This is an approximation — accurate speech rate would require a speech segmentation model to detect pause/speech boundaries. In a production system, this would be replaced with a proper voice activity detection (VAD) algorithm.

4. AI Scoring Model

4.1 Demo Model Architecture

The scoring model in this POC is a weighted heuristic model based on established acoustic correlates of depression documented in clinical research literature. It is NOT a trained machine learning model — it is a rule-based approximation designed to demonstrate the concept for a live demo.

Research Basis

Multiple peer-reviewed studies (including DAIC-WOZ dataset analyses) have identified measurable acoustic features correlated with depressive episodes: reduced pitch mean, reduced pitch range (monotone), lower speech loudness, longer pause durations, slower speech rate, and reduced speaking time ratio.

4.2 Scoring Algorithm

The heuristic scoring function takes three inputs — average dB, average pitch (Hz), and recording duration — and computes a depression likelihood score from 0.0 (no indicators) to 1.0 (strong indicators):

```
baseScore = 0.28
if avgDb < -25: score += 0.22 // Very quiet speech
elif avgDb < -18: score += 0.10 // Moderately quiet
if avgPitch < 150: score += 0.18 // Low / monotone pitch
elif avgPitch < 170: score += 0.08 // Below-average pitch
if duration < 5s: score += 0.06 // Sample too short
if avgDb > -8: score -= 0.10 // Loud / energetic voice
if avgPitch > 210: score -= 0.10 // High / variable pitch
score = clamp(score, 0.04, 0.95)
```

4.3 Interpretation of Results

Score Range	Interpretation	Verdict Display	Action
0–35%	No significant indicators	Green banner	Voice patterns within normal range
35–50%	Marginal indicators	Amber scoring	Monitor over time if concerned
50–65%	Moderate indicators detected	Red banner (warn)	Recommend professional consultation
65%+	Strong acoustic indicators	Red banner (warn)	Recommend urgent professional consultation

4.4 Production Model Options

To replace the heuristic with a real ML model for a production system, the following approaches are recommended:

- Fine-tuned wav2vec2 (HuggingFace) — a pre-trained speech transformer model fine-tuned on depression datasets such as DAIC-WOZ or AVEC. Achieves state-of-the-art performance (~75% accuracy on benchmarks).
- CNN over MFCC features — extract Mel-Frequency Cepstral Coefficients from 30-second audio clips using librosa (Python) and pass through a convolutional neural network trained on labelled data.

- OpenSMILE feature set — an established audio feature extraction toolkit used in clinical research that produces 6,373 acoustic features per clip, suitable for training SVM or gradient boosting classifiers.
- ONNX.js — any of the above models can be exported to ONNX format and run directly in the browser using ONNX.js, maintaining the zero-server architecture of this POC.

5. Hardware & Software Requirements

5.1 Minimum Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Any dual-core CPU (2008 or later)	Quad-core Intel / Apple Silicon / AMD Ryzen
Memory (RAM)	2 GB	4 GB or more
Storage	< 15 KB (single HTML file)	< 15 KB
Microphone	Any built-in or external microphone	USB condenser microphone for cleaner signal
Display	800×600 resolution minimum	1280×720 or higher
Internet	Only required for Google Fonts (DM Sans)	Not required — works fully offline

5.2 Minimum Software Requirements

Platform	Operating System	Browser	Minimum Browser Version
PC / Laptop	Windows 10/11	Chrome, Edge, Firefox	Chrome 74+, Edge 79+, Firefox 76+
PC / Laptop	macOS 11 (Big Sur) or later	Chrome, Safari, Firefox	Safari 14.1+, Chrome 74+
PC / Laptop	Ubuntu 20.04 / Debian 11+	Chrome, Firefox	Chrome 74+, Firefox 76+
Mobile	Android 8.0 (Oreo) or later	Chrome for Android	Chrome 74+
Mobile	iOS 14.5 or later	Safari	Safari 14.5+ (getUserMedia requires iOS 14.5+)
Tablet	iPadOS 14.5 or later	Safari	Safari 14.5+

5.3 Required Browser Permissions

Permission	Why Required	What Happens if Denied
Microphone access	Required to capture voice audio via <code>getUserMedia()</code>	App automatically switches to demo simulation mode — no functionality lost for demo purposes
HTTPS or localhost	<code>getUserMedia()</code> requires a secure context	On local file open (<code>file://</code>), most browsers still allow mic access. Deploy via HTTPS if serving remotely.
JavaScript enabled	All application logic runs in JS	Application will not function

5.4 No Installation Required

Zero Installation

The application requires no software installation, no runtime environment (Node.js, Python, Java), no package manager, no database, no web server, and no internet connection after the file is downloaded. It is a single .html file that runs by double-clicking it in any file manager.

6. How It Works — Step by Step

6.1 Real-Mode Flow (Live Microphone)

Step	What Happens	Technology
1. User clicks Start	Browser prompts for microphone permission	MediaDevices.getUserMedia()
2. Permission granted	Audio stream created with noiseSuppression, echoCancellation, autoGainControl ON	WebRTC audio constraints
3. Audio graph created	MediaStreamSource connected to AnalyserNode inside AudioContext	Web Audio API
4. Waveform rendering	requestAnimationFrame loop reads time-domain buffer every 16ms, draws sine-like trace on Canvas	AnalyserNode + Canvas 2D
5. Feature extraction	Each frame: RMS dB computed from waveform; peak FFT bin computed for pitch Hz	Uint8Array math
6. Metrics update	dB, Hz, wpm~ values update live on screen every animation frame	DOM manipulation
7. Noise bar animates	Progress bar increments to 94% simulating the ongoing noise suppression pipeline	CSS transition + JS interval
8. User clicks Stop	MediaStream tracks stopped; AudioContext suspended; animation cancelled	MediaStreamTrack.stop()
9. Analysis runs	Average dB and pitch computed from all collected samples over the recording duration	Array reduce + math
10. Score computed	Heuristic model applies weighted rules to dB + pitch + duration → score 0–1	JavaScript scoring function
11. Result displayed	Result card animates in with score bars, feature breakdown, and colour-coded verdict	CSS animation + DOM

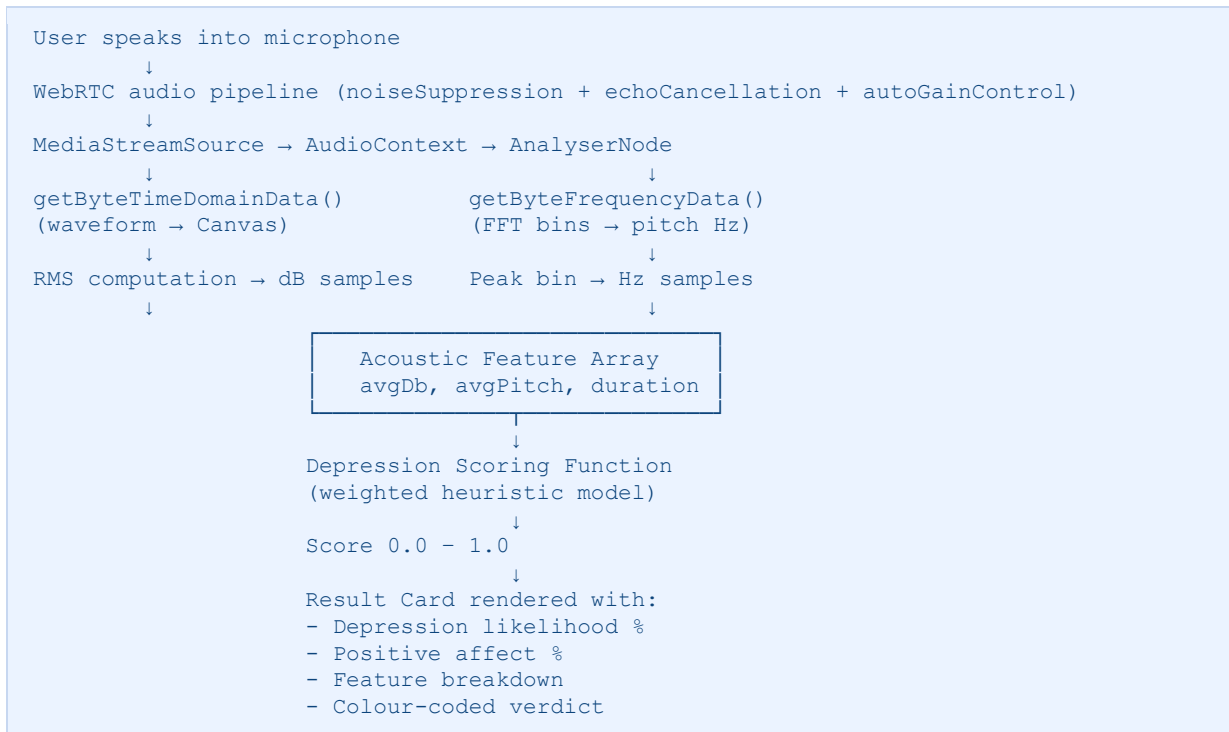
6.2 Demo Simulation Mode Flow

Demo mode runs identically to real mode except audio data is procedurally generated in JavaScript using sine wave mathematics rather than captured from a microphone. This produces realistic-looking waveforms and acoustic metrics without any hardware dependency.

Step	Demo Mode Behaviour
Audio capture	No microphone requested. Uint8Array populated with: $128 + \sin(t + i \times 0.035) \times 22 + \text{noise}$
Pitch simulation	Simulated pitch 148–170 Hz range — slightly below normal to demonstrate a mild depression indicator

Loudness simulation	Simulated dB -20 to -24 range — quiet speech, another mild indicator
Duration	Runs for exactly 10 seconds then auto-stops and analyses
Result	Typically scores ~35–45% — borderline/marginal — suitable for demo discussion

6.3 Data Flow Diagram (Textual)



7. Limitations & Important Disclaimers

CRITICAL DISCLAIMER

This application is a proof-of-concept demonstration tool. It is NOT a medical device, NOT a clinical screening tool, and NOT a diagnostic instrument. The acoustic analysis results must not be used to make any clinical, therapeutic, or personal decisions. Depression is a complex clinical condition that can only be properly assessed by a qualified mental health professional using validated clinical instruments (e.g. PHQ-9, HDRS, MADRS).

7.1 Technical Limitations

- The scoring model is a heuristic approximation, not a trained ML model. It has not been validated against clinical data.
- Pitch detection using peak FFT bins is a coarse approximation. Accurate F0 extraction requires YIN, PYIN, or autocorrelation algorithms.
- A 10–15 second sample is insufficient for reliable acoustic analysis. Research studies typically use 2–5 minute speech samples.
- Background noise, recording environment, accent, language, and individual vocal characteristics significantly affect acoustic features.
- The application has no memory — each session is analysed in isolation with no trend tracking over time.
- Speech rate estimation is a rough proxy based on amplitude — not actual word count detection.

7.2 Clinical Limitations

- Depression presents heterogeneously — not all depressed individuals show vocal acoustic changes.
- Many non-depression conditions (laryngitis, fatigue, anxiety, medication effects) produce similar acoustic patterns.
- Culturally and individually, baseline vocal characteristics vary enormously — a low-pitched voice is not inherently a depression indicator.
- The application has no mechanism to assess suicidal ideation, functional impairment, symptom duration, or other clinical criteria required for diagnosis.

8. Deployment & Usage Instructions

8.1 Running Locally

3. Download the file VoiceDepressionDetector.html to your computer.
4. Double-click the file. It will open in your default web browser.
5. If the browser blocks microphone access for file:// URLs, right-click the file and select Open With, then choose Chrome or Edge.
6. Click Start recording and allow microphone permission when prompted.
7. Speak naturally for 10–20 seconds then click Stop & analyse.
8. The result card will display with your acoustic analysis.

8.2 Demo / Presentation Mode

9. Open the file in Chrome or Edge on the presentation laptop.
10. Click Run demo — no microphone permission required.
11. The app runs a 10-second simulation, shows a live animated waveform, updates metrics, and produces a result — ideal for projector demos.
12. After the demo completes, click Start recording to show real-voice mode.

8.3 Mobile Usage

13. Transfer the HTML file to the mobile device via email, WhatsApp, Google Drive, or USB.
14. On Android: open the file in Chrome. On iOS: open the file in Safari (iOS 14.5+ required for microphone access).
15. Tap the ring button or Start recording. Allow microphone permission. Speak and stop.

8.4 Hosting on a Web Server (Optional)

If you want to share the app via a URL rather than a file, upload VoiceDepressionDetector.html to any static web host:

- GitHub Pages (free) — create a repository, upload the file, enable Pages. App runs at <https://username.github.io/repo/VoiceDepressionDetector.html>
- Netlify Drop (free) — drag and drop the file at app.netlify.com/drop. Instant public URL.
- Vercel / Cloudflare Pages (free) — deploy via CLI or UI for a permanent URL.
- Local Python server — run: `python -m http.server 8080` in the file's directory, then open <http://localhost:8080/VoiceDepressionDetector.html>

Hosting on HTTPS is required for microphone access on remote devices. Local file:// works on the same machine.

9. Future Enhancement Roadmap

Priority	Enhancement	Technology / Approach
High	Replace heuristic with trained ML model (wav2vec2)	HuggingFace Transformers → ONNX.js for browser inference
High	Accurate pitch detection (YIN / PYIN algorithm)	JavaScript implementation of autocorrelation-based F0 detection
High	MFCC feature extraction in browser	Custom JS FFT + mel filterbank, or TensorFlow.js DSP library
Medium	Session history and trend tracking	localStorage / IndexedDB for multi-session acoustic trend analysis
Medium	PHQ-9 questionnaire integration	Combine acoustic score with validated clinical questionnaire score
Medium	Speaker diarisation	Identify who is speaking in a multi-person conversation
Medium	Native mobile app (iOS/Android)	React Native or Flutter wrapping the same Web Audio pipeline
Low	Backend API with clinical-grade model	Python FastAPI + librosa + OpenSMILE feature extraction
Low	Longitudinal monitoring dashboard	Track acoustic biomarkers over weeks/months with trend visualisation
Low	Multi-language support	Language-specific acoustic baseline models for non-English speakers

10. References & Further Reading

- Cummins, N. et al. (2015). A review of depression and suicide risk assessment using speech analysis. *Speech Communication*, 71, 10–49.
- Gratch, J. et al. (2014). The Distress Analysis Interview Corpus of Human and Computer Interviews (DAIC-WOZ dataset). LREC.
- W3C Web Audio API Specification — <https://www.w3.org/TR/webaudio/>
- W3C Media Capture and Streams — <https://www.w3.org/TR/mediacapture-streams/>
- WebRTC Audio Processing — <https://webrtc.org/getting-started/media-capture-and-constraints>
- HuggingFace depression detection models — <https://huggingface.co/models?search=depression+speech>
- OpenSMILE acoustic feature toolkit — <https://audeering.github.io/opensmile/>
- ONNX.js browser inference — <https://github.com/microsoft/onnxruntime/tree/main/js/web>
- PHQ-9 Depression Screening Tool — Kroenke, K. et al. (2001). *Journal of General Internal Medicine*.

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